

LAB 1

1. Write a C# program to display "Hello, World!" on the screen.

```
class Program
{
    static void Main(string[] args)
    {
        Console.WriteLine("Hello World!");
    }
}
```

```
yanil@fedora:~/Desktop/6th_sem/DotNet/lab/Lab1$ dotnet run
Hello World!
```

2. Write a C# program to declare and print variables of different data types: int, double, char, bool, and string.

```
class Program
{
    static void Main(string[] args)
    {
        int rollNo = 101;
        double cgpa = 3.5;
        string name = "Rahul";
        char gender = 'M';
        bool isPass = true;
        Console.WriteLine("Roll no. : " + rollNo);
        Console.WriteLine("Name : " + name);
        Console.WriteLine("Gender : " + gender);
        Console.WriteLine("CGPA : " + cgpa);
        Console.WriteLine("Passed : " + isPass);

    }
}
```

```
yanil@fedora:~/Desktop/6th_sem/DotNet/lab/Lab1$ dotnet run
Roll no. : 101
Name : Rahul
Gender : M
CGPA : 3.5
Passed : True
```

3. Write a C# program to perform basic arithmetic operations using two numbers.

```
class Program
{
    static void Main(string[] args)
```

```

{
    int num1 = 40;
    int num2 = 20;
    Console.WriteLine("Addition : " + (num1 + num2));
    Console.WriteLine("Subtraction : " + (num1 - num2));
    Console.WriteLine("Multiplication : " + (num1 * num2));
    Console.WriteLine("Division : " + (num1 / num2));
}
}

```

```

yanil@fedora:~/Desktop/6th_sem/DotNet/lab/Lab1$ dotnet run
Addition : 60
Subtraction : 20
Multiplication : 800
Division : 2

```

4. Write a C# program to check whether a given number is even or odd using an if-else statement.

```

class Program
{
    static void Main(string[] args)
    {
        int num = 27;
        if (num % 2 == 0)
            Console.WriteLine("Given number is Even");
        else
            Console.WriteLine("Given number is Odd");
    }
}

```

```

yanil@fedora:~/Desktop/6th_sem/DotNet/lab/Lab1$ dotnet run
Given number is Odd

```

5. Write a C# program to find the largest number among three numbers using nested if-else statement.

```

class Program
{
    static void Main(string[] args)
    {
        int num1 = 23, num2 = 12, num3 = 20;
        int largest;
        if (num1 > num2 && num1 > num3)
            largest = num1;
        else if (num2 > num1 && num2 > num3)

```

```

        largest = num2;
    else largest = num3;
    Console.WriteLine("Largest number : "+largest);
}
}

```

```

yanil@fedora:~/Desktop/6th_sem/DotNet/lab/Lab1$ dotnet run
Largest number : 23

```

6. Write a C# program to display the first 10 natural numbers using a for-loop.

```

class Program
{
    static void Main(string[] args)
    {
        Console.Write("Natural numbers: ");
        for (int i = 1; i <= 10; i++)
        {
            Console.Write(" " + i);
        }
        Console.WriteLine();
    }
}

```

```

yanil@fedora:~/Desktop/6th_sem/DotNet/lab/Lab1$ dotnet run
Natural numbers:  1 2 3 4 5 6 7 8 9 10

```

7. Write a C# program to print the multiplication table of a given number using while loop.

```

class Program
{
    static void Main(string[] args)
    {
        Console.WriteLine("Multiplication Table of 5 : ");
        int num=5, i=1;
        while (i <= 10)
        {
            Console.WriteLine(num+"x"+i+"="+num*i);
            i++;
        }
    }
}

```

```
yanil@fedora:~/Desktop/6th_sem/DotNet/lab/Lab1$ dotnet run
Multiplication Table of 5 :
5x1=5
5x2=10
5x3=15
5x4=20
5x5=25
5x6=30
5x7=35
5x8=40
5x9=45
5x10=50
```

8. Write a C# program to display the first 10 natural numbers in reverse order using do-while loop.

```
class Program
{
    static void Main(string[] args)
    {
        Console.Write("Natural numbers in reverse order : ");
        int i = 10;
        do
        {
            Console.Write(" " + i);
            i--;
        } while (i >= 1);
        Console.WriteLine();
    }
}
```

```
yanil@fedora:~/Desktop/6th_sem/DotNet/lab/Lab1$ dotnet run
Natural numbers in reverse order : 10 9 8 7 6 5 4 3 2 1
```

9. Write a C# program to store 5 numbers in an array and display them.

```
class Program
{
    static void Main(string[] args)
    {
        int[] numbers = { 13, 21, 23, 14, 25 };
        for (int i = 0; i < numbers.Length; i++)
        {
            Console.Write( numbers[i] + " ");
        }
        Console.WriteLine();
    }
}
```

```
yanil@fedora:~/Desktop/6th_sem/DotNet/lab/Lab1$ dotnet run
13 21 23 14 25
```

10. Write a C# program to find the sum of elements of an array.

```
class Program
{
    static void Main(string[] args)
    {
        int[] numbers = { 13, 21, 23, 14, 25 };
        int sum = 0;
        foreach (var num in numbers)
        {
            sum += num;
        }
        Console.WriteLine("Sum of array : " + sum );
    }
}
```

```
yanil@fedora:~/Desktop/6th_sem/DotNet/lab/Lab1$ dotnet run
Sum of array : 96
```